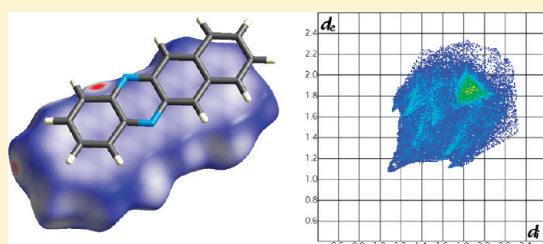


Acenes vs N-Heteroacenes: The Effect of N-Substitution on the Structural Features of Crystals of Polycyclic Aromatic Hydrocarbons

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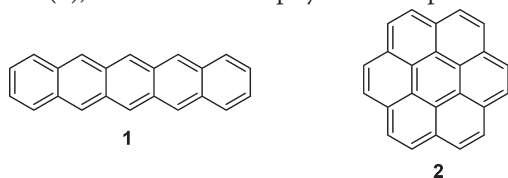
ABSTRACT: Effective charge transport in organic semiconductors requires orbital overlap between neighboring molecules and, therefore, is strongly dependent on solid state packing. Acenes such as pentacene, which have been widely studied as organic semiconductors, exhibit solid state packing motifs that are dominated by C–H··· π interactions and show very little π -stacking. Recently, N-heteroacenes have emerged as potential alternatives to acenes as semiconducting materials. These compounds have altered frontier molecular orbital energies, show improved stability as compared to their hydrocarbon analogues, and are readily synthesized. However, the solid state structural features have not been systematically examined. In this study, the crystal structures of anthracene, tetracene, and pentacene are compared to N-substituted analogues using Hirshfeld surface analysis. While the acenes exhibit herringbone packing that is dominated by C–H··· π interactions, the N-heteroacenes exhibit distinctly different packing motifs featuring π -stacking interactions and C–H···N interactions. This trend is also observed when other hydrocarbons such as phenanthrene and chrysene are compared to N-heterocyclic analogues. In contrast, the dihydrodiazacenes that are produced by reduction of the N-heteroacenes exhibit herringbone packing motifs that are similar to those of their hydrocarbon analogues. These differences in packing are attributed to differences in electrostatics between the electron-rich acenes and the electron-deficient N-heteroacenes.



■ INTRODUCTION

Polycyclic aromatic hydrocarbons (PAHs) have been widely studied in part because their semiconducting properties make them potentially useful for applications in organic electronics.^{1–4} In particular, acenes such as pentacene have been widely studied for their semiconducting properties, demonstrating excellent charge transport properties.^{5,6} Charge transport in organic semiconductors depends not only on molecular properties such as frontier orbital energies and reorganization energy, but also on the electronic coupling between molecules. Direct cofacial π -stacking interactions are thought to improve orbital overlap, which in turn improves the electronic coupling that is essential for efficient charge transport.^{7,8} Consequently, an understanding of the solid state organization of PAHs is important for the development of new materials where π -stacking is desired.

As a result of the pioneering studies of Desiraju and Gavezzotti and subsequent work, the patterns of solid state organization in crystal structures of PAHs are well understood.^{9–11} The crystal structures of PAHs feature C–H··· π interactions and π – π stacking interactions in varying degrees, depending on molecular structure. For PAHs where the C/H ratio is relatively low, including acenes such as in pentacene (**1**), C–H··· π interactions dominate molecular association, leading to a herringbone packing motif.^{12,13} For PAHs with a higher C/H ratio, such as coronene (**2**), π – π interactions play a more important role.



More recently, attention has turned to the properties of N-substituted heteroacenes as potential semiconducting materials.^{14–20} N-Heteroacenes have significantly different frontier molecular orbital energies as compared with their hydrocarbon analogues, and they have been proposed as potential n-type semiconductors.¹⁸ Specifically, some diazatetracenes have high electron affinities and less negative reduction potentials, suggesting that they may be good candidates as electron accepting materials.²⁰ Houk and Winkler have also suggested that N-heteroacenes may self-organize into sheets via C–H···N interactions.¹⁸ Similarly, dihydrodiazacenes have been proposed as isostructural alternatives to acenes, owing to their relative ease of synthesis, improved stability, and solubility.^{21,22} Since charge transport properties in organic semiconductors depend on solid state organization, an understanding of intermolecular interactions in crystals of heteroacenes is important for assessing their utility as new materials for organic electronics. Although the crystal structures for several heteroacenes have been reported, their solid state packing has not been described in detail, and the effect of N-substitution on solid state organization of acenes has not been systematically examined. Given the recent interest in N-heteroacenes as novel materials for organic electronics, there is considerable impetus for understanding the molecular packing features in this class of compounds.

Herein the structural features of crystals of acenes and their N-heterocyclic analogues are systematically compared. Simple acenes and N-heterocyclic analogues have been chosen to

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provide a comparison of solid state packing in the absence of the effects of other substituents. Intermolecular interactions are evaluated by analysis of Hirshfeld surfaces. Hirshfeld surfaces have recently emerged as a powerful tool for describing intermolecular interactions in crystals.^{23–28} Fingerprint plots generated from Hirshfeld surfaces allow the closest intermolecular contacts to be separated and their relative proportions to be quantified, thereby providing an indication of the relative contributions of different types of intermolecular interactions. Hirshfeld surface analysis is particularly useful for identifying differences in the packing motifs of aromatic compounds, notably PAHs^{11,29} and related compounds bearing multiple phenyl groups.^{30,31} Analysis of the Hirshfeld surfaces provides a relative proportion of C···H intermolecular contacts that are attributed predominantly to C–H··· π interactions and

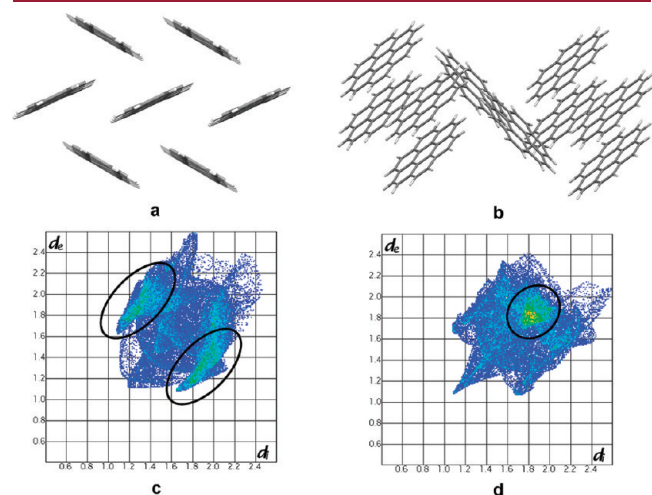


Figure 1. (a) View of the crystal structure of pentacene (1); (b) view of the crystal structure of coronene (2); (c) Hirshfeld 2D fingerprint plot of 1 with the regions of the plot corresponding to C···H contacts circled; (d) Hirshfeld 2D fingerprint plot of 2 with the region of the plot corresponding to C···C contacts circled.

C···C interactions that reflect π -stacking interactions. Hirshfeld surfaces can also be used to generate 2D “fingerprint” plots, where the distance from the surface to the nearest interior atom (d_i) is plotted against the distance from the surface to the nearest exterior atom (d_e).²² In the resulting graphs, different regions of the plot represent different types of closest intermolecular contacts (Figure 1). For example, the Hirshfeld 2D fingerprint plot of pentacene (1), which exhibits a herringbone packing motif dominated by C–H··· π interactions, has the closest intermolecular contacts of 0.4% for C···C, 58.6% for C···H, and 41% for H···H contacts (Figure 1). In contrast, coronene (2) exhibits a γ -packing motif with significant π -stacking interactions. Hirshfeld analysis of 2 shows 19.5% C···C contacts, 30.7% C···H contacts, and 49.7% H···H contacts.

The systematic comparison of N-heteroacenes with their hydrocarbon analogues presented here shows that the two classes of compounds display distinctly different packing motifs, despite their similar molecular topologies. The results of this study should provide crystal engineers with additional design tools for controlling organization in crystals of aromatic compounds and have implications for the use of these compounds as semiconducting materials.

EXPERIMENTAL SECTION

Crystal structures were obtained from the Cambridge Structural Database using Conquest 1.13. The following filters were applied to the structure searches: 3D coordinates determined, not disordered, no errors, not polymeric, no ions, no powder structures, and only organics. In cases where multiple reports were found, a representative structure with a low *R* factor was chosen. Where polymorphs were found, they were analyzed separately and representative examples presented. Hirshfeld surface analysis was conducted using CrystalExplorer.³²

RESULTS AND DISCUSSION

The crystal structures of anthracene (3), tetracene (4), and pentacene (1) show characteristic herringbone packing,

Table 1. Intermolecular Contacts Determined by Analysis of Hirshfeld Surfaces for Acenes and Their N-Heterocyclic Analogues

compd	CSD code	% C···C	% C···H	% H···H	% C···N	% N···H	% N···N
anthracene (3)	ANTCEN	1.8	49.1	49.1			
tetracene (4)	TETCEN	0.7	56.8	43.3			
pentacene (1)	PENCEN	0.4	58.6	41			
acridine (form II) (5)	ACRIDIN04	7.5	33	49.4	2.8	7.3	0
acridine (form III) (5)	ACRIDIN07	7.7	35.8	44.6	2.3	9.6	0
phenazine (form α) (6)	PHENAZ04	8.3	27.1	44.5	4.8	15.3	0
phenazine (form β) (6)	PHENAZ11	7.5	29.1	44.4	2.6	16.2	0.2
diazatetracene (7)	LODCOX	14.5	21.4	48	4	11.7	0.4
phenanthrene (8)	PHENAN04	0.4	51.5	48.2			
phenanthridine (9)	PHENAT02	11.1	28.7	50	1.6	8.6	0
<i>o</i> -phenanthroline (10)	OPENAN	8.8	37.4	36.3	0.2	17.1	0.1
1,7-phenanthroline (11)	QARYEO	7.7	28.2	44.4	3.1	16.3	0.3
9,10-diazaphenanthrene (12)	DAZPNE	6.3	30.9	40.4	4	18.4	0
chrysene (13)	CRYSEN	0.4	58	41.6			
benzo[<i>i</i>]phenanthridine (14)	OFODUJ	6.8	39.6	44.2	2.4	7.1	0
dihydrophenazine (15)	MEMTED	0	45.8	45.7	0	8.5	0
dihydrodiazatetracene (16)	LOJMAZ	0.2	57.3	42.5	0	0	0
dihydrodiazapentacene (17)	VANXEO	0	55.3	39.6	0	5.1	0

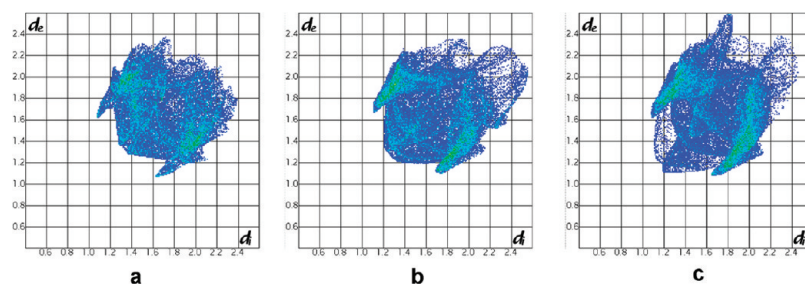
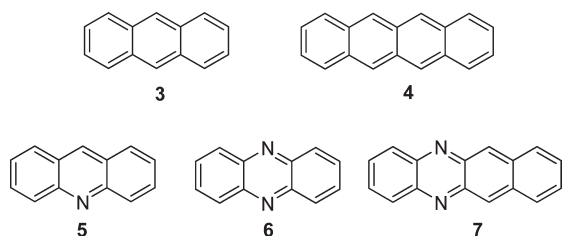


Figure 2. Representative Hirshfeld 2D fingerprint plots of (a) 3 (ANTCEN), (b) 5 form II (ACRDIN04), and (c) 6 form α (PHENAZ04).

featuring prominent $\text{C}-\text{H}\cdots\pi$ interactions (edge-to-face interactions) and virtually no direct $\pi-\pi$ interactions (Figure 1a). Hirshfeld surface analysis of anthracene reveals a relatively large proportion of $\text{C}\cdots\text{H}$ contacts (49.1%) and only 1.8% $\text{C}\cdots\text{C}$ contacts (Table 1). The relative importance of $\text{C}\cdots\text{H}$ contacts increases with increasing acene length, due to the decreased C/H ratio and the increased number of C–H groups available to engage in these interactions.



Acridine (5), where one CH unit of anthracene is replaced by a nitrogen atom, has a very similar shape to the parent hydrocarbon and might be expected to show similar packing in the solid state. There are several known polymorphs of acridine: form I is a hydrate, form II is the most stable polymorph, and form III is also accessible under ambient conditions.^{33–37} As representative examples, forms II and III are discussed here. Hirshfeld analysis of both of these polymorphs shows that they exhibit very different solid state organization compared to anthracene (Figure 2), as evidenced by a significantly lower proportion of $\text{C}\cdots\text{H}$ contacts (33% and 35.8% for forms II and III, respectively, compared to 49.1% for 3) and an increase in the percentage of $\text{C}\cdots\text{C}$ contacts (7.5% and 7.7%, respectively, compared to 1.8% for 3).

Form II of 5 is characterized by $\text{C}-\text{H}\cdots\text{N}$ dimer interactions between two neighboring molecules ($\text{N}\cdots\text{H}$ distance = 2.783 Å), as shown in Figure 3a. In addition, each molecule exhibits an offset π -stacking interaction with another neighbor (the two closest centroid–centroid distances are 3.67 Å). Form III also exhibits a $\text{C}-\text{H}\cdots\text{N}$ dimer motif ($\text{N}\cdots\text{H}$ distance = 2.725 Å) as well as limited offset π -stacking (centroid–centroid distance of 3.696 Å) where one of the terminal rings overlaps with the terminal ring of a neighboring molecule.

Phenazine (6), where two CH units have been replaced by nitrogen, has been reported as two polymorphs (forms α and β).^{38–40} Form α (PHENAZ04) features a relatively lower proportion of $\text{C}\cdots\text{H}$ (27.1%) contacts associated with edge-to-face $\text{C}-\text{H}\cdots\pi$ interactions as compared with anthracene and a higher proportion of $\text{C}\cdots\text{C}$ contacts associated with π -stacking (8.3%). In addition, as the number of nitrogen atoms is increased (a higher N/H ratio), $\text{N}\cdots\text{H}$ contacts become more prevalent (15.3% N–H contacts). The $\text{N}\cdots\text{H}$ contacts are

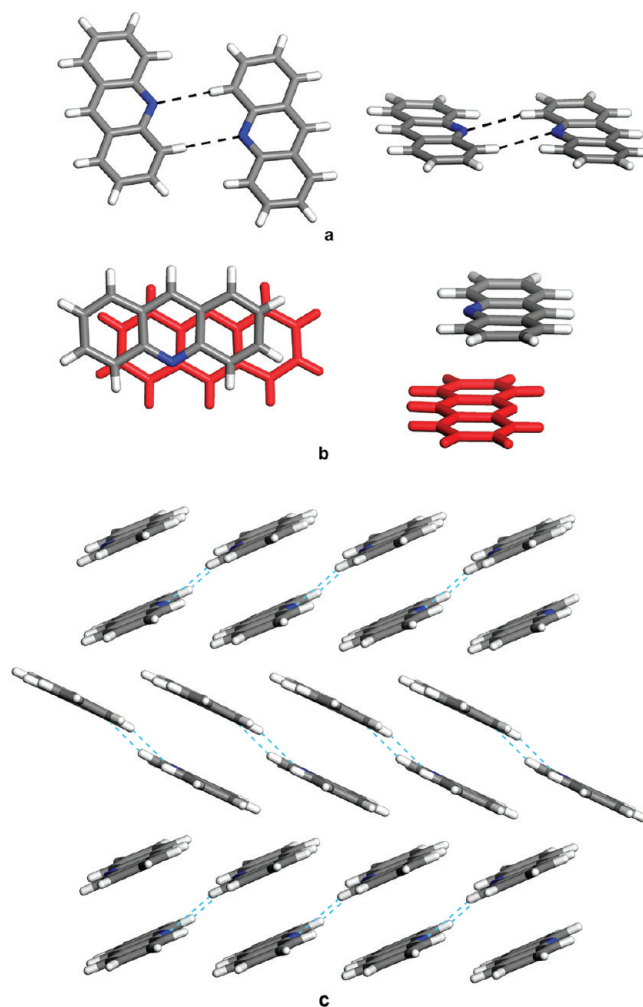


Figure 3. Representations of the crystal structure of 5 (ACRDIN04) showing (a) the $\text{C}-\text{H}\cdots\text{N}$ dimer motif, (b) the offset π -stacking to form a dimer, and (c) the overall packing.

manifested as $\text{C}-\text{H}\cdots\text{N}$ dimer interactions ($\text{N}\cdots\text{H}$ distance = 2.893 Å), where each molecule participates in this characteristic dimer with two neighboring molecules, forming an extended 1D chain along the a -axis (Figure 4). These 1D chains interact with adjacent chains via offset $\pi-\pi$ interactions (centroid–centroid distances = 3.817 Å) to form an extended π -stacked array.

Hirshfeld surface analysis of form β of 6 (PHENAZ11) reveals similar intermolecular contacts as compared with the α polymorph. Again, characteristic $\text{C}-\text{H}\cdots\text{N}$ dimer interactions that form an extended 1D chain are observed ($\text{N}\cdots\text{H}$ distances =

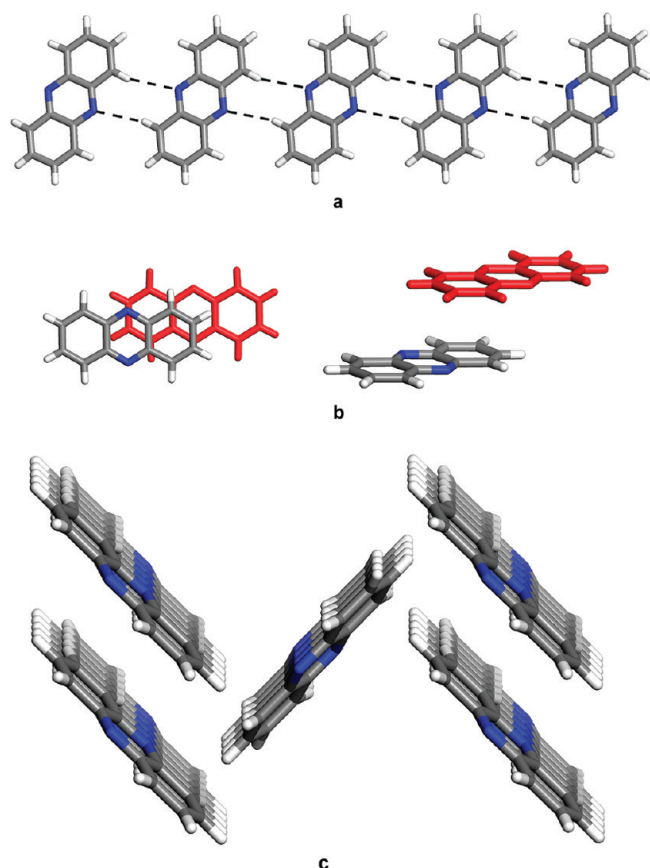


Figure 4. Representations of the crystal structure of **6** (PHENAZO4) showing (a) the C–H···N interactions to form a 1D chain, (b) the offset π -stacking to form a dimer, and (c) the overall packing.

2.708, 2.709 Å), as are offset π -stacking interactions (closest centroid–centroid distances of 3.712 Å). In this polymorph, the π -stacking does not form an extended array; rather, it is confined to stacked dimers.

Clearly, replacement of CH units of anthracene with nitrogen atoms has a significant impact on solid state organization, despite the minimal change in molecular shape. While the parent hydrocarbon has a crystal structure that is dominated by C–H··· π interactions that result in herringbone packing, the N-heterocyclic analogues are characterized by reduced C···H contacts, as shown by Hirshfeld surface analysis. The structures of the N-heterocyclic analogues of anthracene exhibit characteristic C–H···N dimer interactions and an increase in the relative proportions of C···C contacts that are the result of offset π -stacking interactions.

To further probe the effects of N-substitution on the packing motifs of acenes, the crystal structures of tetracene⁴¹ (**4**) and diazatetracene²⁰ (**7**) were also examined using Hirshfeld surface analysis. Comparison of the 2D fingerprint plots of **4** and **7** showed even more dramatic differences in packing (Figure 5). While the structure of **4** is dominated by C···H contacts (56.2%) with virtually no C···C contacts (0.5%), the structure of **7** shows a dramatically lower proportion of C···H contacts (21.4%) and a significantly higher proportion of C···C contacts (14.5%).

The structure is characterized by extended π -stacks with an average distance between molecular planes of approximately

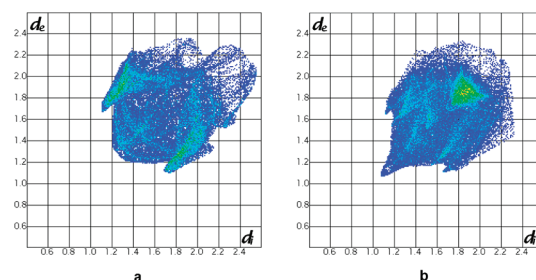
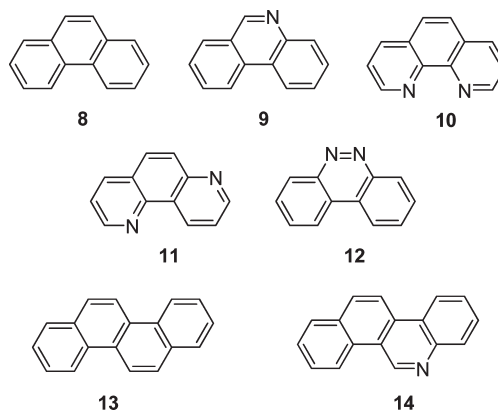


Figure 5. 2D fingerprint plots of (a) **4** (TETCEN) and (b) **7** (LODCOX).

3.51 Å. The π -stacks consist of dimers that are almost directly overlapped but which interact with neighboring dimers in an offset manner (Figure 6). These “staggered” π -stacked arrays in turn interact with adjacent stacks via C–H··· π interactions involving the short end of the molecule to form a quasi-herringbone packing motif (Figure 6c).

These observations underscore the tendency of N-heteroacenes to engage in π -stacking interactions and a reduced tendency to exhibit C–H··· π interactions. It is anticipated that the crystal structure of the corresponding diazapentacene would display similar features; although diazapentacene has been synthesized,¹⁹ its crystal structure has unfortunately not been reported.

In order to determine whether the effects of N-substitution on solid state organization extend beyond the acene series, other PAHs characterized by extensive C–H··· π interactions were examined. For example, phenanthrene (**8**) also displays classic herringbone packing with significant C···H contacts (51.5%) and little C···C contacts (0.4%). Phenanthridine (**9**), however, shows decreased C···H contacts (28.7%) and increased C···C contacts (11.1%). Other analogues of phenanthrene containing two nitrogen atoms (**10**–**12**) also show increases in the relative importance of C···C and N···H contacts at the expense of C···H contacts. Similarly, chrysene (**13**) packs in a herringbone motif and has 0.4% C···C contacts, 58% C···H contacts, and 41.6% H···H contacts. Hirshfeld surface analysis of benzo-[i]phenanthridine (**14**) gives 6.8% C···C contacts, 39.6% C···H contacts, 44.2% H···H contacts, and 7.1% N···H contacts.



The crystal structures considered in this study consistently show that solid state organization of N-heteroacenes and related compounds is characterized by C–H···N interactions, a reduced tendency to engage in C–H··· π interactions, and an

increased propensity to engage in π -stacking interactions as compared to their hydrocarbon analogues.

While the crystal structure of diazapentacene has not been reported, the structure of dihydrodiazapentacene has been reported to be comparable to pentacene itself,²¹ suggesting that reduction of the diazaacenes to their corresponding dihydro- analogues results in packing similar to that of the parent acenes. Hirshfeld surface analysis of dihydrophenazine

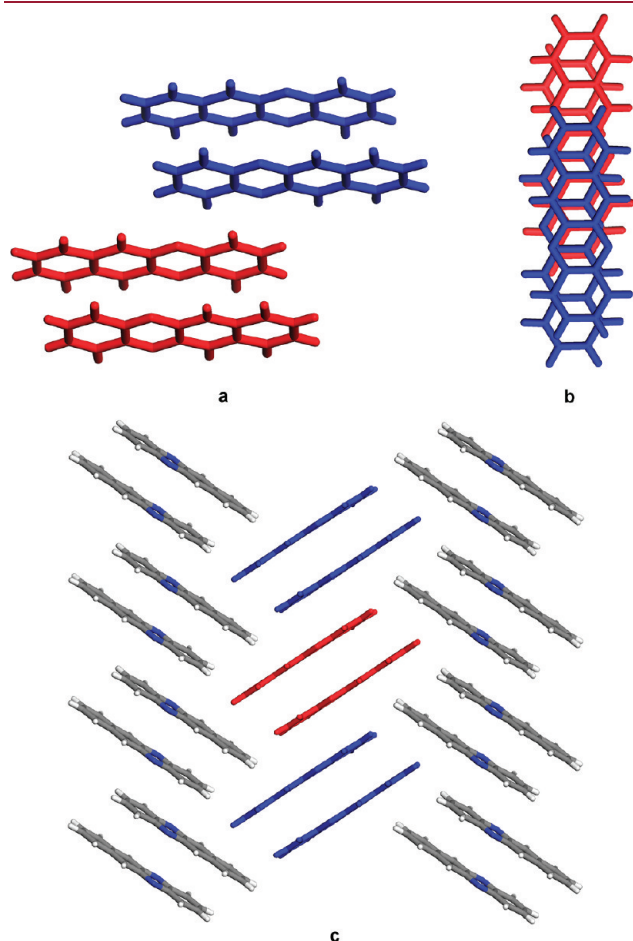
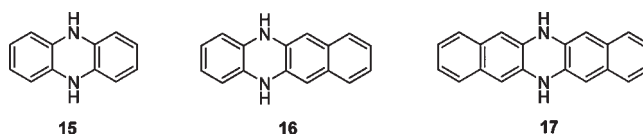


Figure 6. Representations of the crystal structure of **7** (LODCOX), showing (a) four molecules consisting of two adjacent π -stacked dimers in red and blue, (b) these dimers viewed down the π -stacking axis, and (c) the organization of these dimers into a quasi-herringbone arrangement with $C-H\cdots\pi$ interactions involving the short ends of the molecules.

(**15**), dihydrodiazatetracene (**16**), and dihydrodiazapentacene (**17**) all show fingerprint plots that are similar to their hydrocarbon analogues (Figure 7), with a high proportion of $C\cdots H$ contacts and very low proportion of $C\cdots C$ contacts. Furthermore, like the parent acenes, compounds **16** and **17** exhibit a herringbone packing motif. Dihydrophenazine (**15**) is not fully planar in the solid state, showing a slight bend at the sp^3 -hybridized nitrogen atom. Nevertheless, the packing of **15** in the solid state is dominated by $C-H\cdots\pi$ interactions and closely resembles a classic herringbone motif.



Overall, N-heterocyclic acenes show molecular packing features that are distinctly different from those of the corresponding hydrocarbons, while the dihydro- analogues show packing that very closely resemble that of the parent acenes. These observed differences in packing between the acenes, diazaacenes, and hydrodiazacenes are consistent with the electrostatic model for arene–arene interactions by Hunter and Saunders.^{21,42} $C-H\cdots\pi$ interactions are expected to be favored in electron-rich aromatic systems such as the parent acenes and the dihydrodiazacenes, resulting in a herringbone packing motif. In contrast, the N-heteroacenes are electron deficient, making $C-H\cdots\pi$ less favorable. Consequently, the N-heteroacenes exhibit distinctly different packing motifs that feature π -stacking interactions and $C-H\cdots N$ interactions. Qualitative support for the role of electrostatics in the solid state organization of these compounds is provided by a comparison of the electrostatic potential maps for anthracene, acridine, phenazine, and dihydrophenazine (Figure 8).

The electrostatic potential maps show that both anthracene and dihydrophenazine are electron-rich π -systems with significant negative electrostatic potentials above and below the aromatic rings. In contrast, acridine and phenazine are electron deficient, having substantially lower negative electrostatic potentials above and below the rings. The herringbone packing motif dominated by $C-H\cdots\pi$ interactions observed for anthracene and dihydrophenazine is consistent with the electrostatic attraction of the electron-poor hydrogen atoms with the electron rich aromatic ring. For acridine and phenazine, on the other hand, such a packing motif is not electrostatically favored, leading to an altered packing arrangement that features $C-H\cdots N$ hydrogen bonds and a larger proportion of $C\cdots C$ contacts consistent with π -stacking.

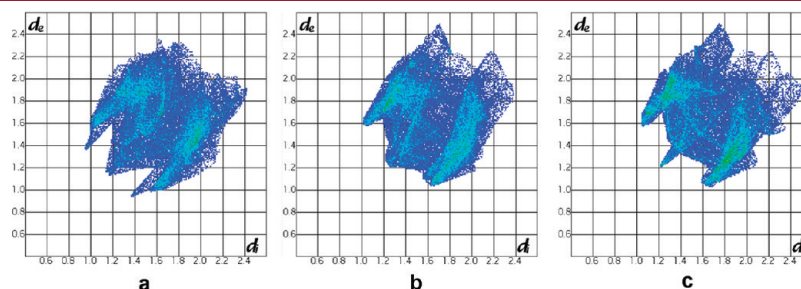


Figure 7. Hirshfeld fingerprint plots of dihydrodiazacenes (a) **15** (MEMTED), (b) **16** (LOJMAZ), and (c) **17** (VANXEO).

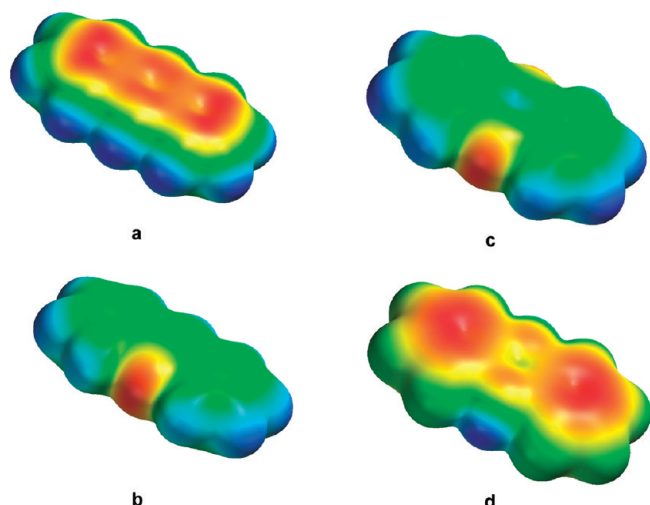


Figure 8. Electrostatic potential maps calculated at the B3LYP/6-31G** level for (a) anthracene (3), (b) acridine (5), (c) phenazine (6), and (d) dihydrophenazine (15).

CONCLUSIONS

The results of this study indicate that heteroatom substitution in PAHs has a significant impact on solid state organization. Specifically, replacement of CH groups in acenes and related compounds by nitrogen atoms leads to crystal structures where the importance of C $\cdots$ H contacts is diminished and the proportion of C $\cdots$ C contacts is greater. Consequently, the N-heteroacenes do not exhibit the typical herringbone packing motif of their hydrocarbon analogues but show structures that feature C–H $\cdots$ N interactions and an increased tendency to engage in π – π stacking interactions. Interestingly, the corresponding dihydrodiazacenes show packing characteristics that are comparable to those of the analogous acenes. These observed differences in packing can be qualitatively attributed to differences in the electrostatic potentials of these compounds. Overall, the results of this study suggest that N-heteroacenes display packing characteristics that may be exploited to produce molecular crystals where π -stacking is favored over C–H $\cdots$ π interactions. Furthermore, the solid state packing of N-heteroacenes may contribute to more effective charge transport due to more effective orbital overlap, which supports their potential utility as alternatives to acenes as organic semiconducting materials.

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